Probability Models & Axioms

Sample Space & Axioms

- **Sample Space** (Ω): Outcomes must be mutually exclusive and collectively exhaustive.
- **Axioms**: $\mathbb{P}(A) \geq 0$, $\mathbb{P}(\Omega) = 1$.
  If $A \cap B = \emptyset \implies \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- **Properties**: $\mathbb{P}(\emptyset) = 0$. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
  $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \impliedby \Omega = B \cup B^c$.
  $A \subset B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
  $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- **Union Bound**: $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.
- **Discrete Uniform**: If Ω has n equally likely elements and A has k, $\mathbb{P}(A) = k/n$.
- **De Morgan**: $(\cup S_n)^c = \cap S_n^c$. $(\cap S_n)^c = \cup S_n^c$.
- **Inter. Bound**: $\mathbb{P}(\cap_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbb{P}(A_i) - (n-1)$.

Conditioning & Independence

- **Conditional Prob**:
  $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ for $\mathbb{P}(B) > 0$.
- **Uniform Cond.** $\mathbb{P}$:
  if given B the outcomes are equally likely
  $\mathbb{P}(A|B) = |A \cap B|/|B|$.
- **Multiplication**:
  $\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}(A_i | \cap_{j=1}^{i-1} A_j)$.
- **Total Prob**: For partition A_i,
  $\mathbb{P}(B) = \sum \mathbb{P}(A_i)\mathbb{P}(B|A_i)$.
- **Bayes' Rule**:
  $\mathbb{P}(A_i|B) = \mathbb{P}(A_i)\mathbb{P}(B|A_i) / \sum \mathbb{P}(A_j)\mathbb{P}(B|A_j)$.
- **Independence**:
  $A \perp\!\!\!\perp B \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, also:
  $A \perp\!\!\!\perp B \iff \mathbb{P}(A|B) = \mathbb{P}(A)$, $\quad (\mathbb{P}(B) > 0)$.
  $A \perp\!\!\!\perp B \iff A \perp\!\!\!\perp B^c$.
- **Conditional Independence:** ($\mathbb{P}(C) > 0$):
  $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C)$
  If, in addition, $\mathbb{P}(B \cap C) > 0$, this relationship is equivalent to the condition:
  $$\mathbb{P}(A \mid B \cap C) = \mathbb{P}(A \mid C)$$
- **Independence of Several Events:** **($\neq$ Pairwise.)**
  Events $A_1, A_2, \ldots, A_n$ are independent if:
  $\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i), \quad \forall S \subseteq \{1, 2, \ldots, n\}$
- **Expectation of $X \perp\!\!\!\perp Y$**: $X \perp\!\!\!\perp Y \implies$
  $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.
  $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$.
- **Variance of $X \perp\!\!\!\perp Y$**: $X \perp\!\!\!\perp Y \implies$
  $\mathrm{Var}(aX + bY + c) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 0$.
  Trap: $X \perp\!\!\!\perp Y \implies \mathrm{Cov}(X, Y) = 0$.
  Converse is not necessarily true.

Counting

r-stage process, n_i possible results at the ith stage:

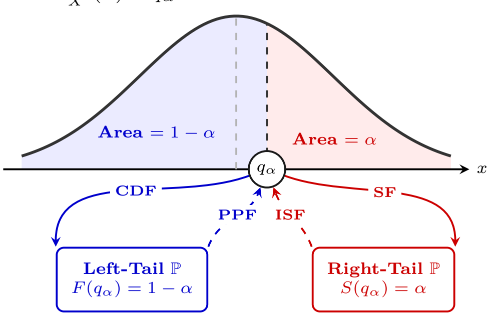
total number of possible results $= n_1 n_2 \cdots n_r$

- **Permutations** of n objects: $n!$
- **k-permutations** of n objects: $\frac{n!}{(n-k)!}$
- **Combinatns** of k out of n objects: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
- **Partitions** of n objects into r groups, ith group having n_i objects. Or subsets of sizes n_i sum to n.:
  $$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$$

Random Variables (RVs)

Discrete & Continuous Basics

- **PMF**: $p_X(x) = \mathbb{P}(X = x)$. With $\sum p_X(x) = 1$.
- **PDF**: $f_X(x) \geq 0$. With $\int f_X(x)dx = 1$.
- **CDF**: $F_X(x) = \mathbb{P}(X \leq x)$, $F_X'(x) = f_X(x)$.
  $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$
- **SF**: $S_X(x) = \mathbb{P}(X > x)$. The "Survival function".
- **PPF**: $F_X^{-1}(1 - \alpha) = q_\alpha$,
- **ISF**: $S_X^{-1}(\alpha) = q_\alpha$.



- **Expectation**: $\mathbb{E}[X] = \sum x p_X(x) = \int x f_X(x) dx$.
- **Expected Value Rule for Func. of RVs**:
  $\mathbb{E}[g(X)] = \sum g(x)p_X(x) = \int g(x) f_X(x)dx$.
- **Linearity**: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$.
- **Variance**:
  $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
  $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$.
  $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$.
- **Covariance**: $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
  $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
- **Correlation**: Location/scale invariant:
  $\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \mathbb{E}[\frac{(X - \mathbb{E}[X])}{\sigma_X} \frac{(Y - \mathbb{E}[Y])}{\sigma_Y}]$
- **Properties**:
  $\mathrm{Cov}(aX, bY + Z) = ab\mathrm{Cov}(X, Y) + a\mathrm{Cov}(X, Z)$.
- **Covariance Matrix (Σ)**: $\mathbf{X} \in \mathbb{R}^k$:
  $\Sigma = \mathbb{E}\left[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T\right]$
  $\Sigma = \mathrm{Var}(\mathbf{X}) = \mathrm{Cov}(\mathbf{X}, \mathbf{X}) =$ "$\mathrm{Cov}(\mathbf{X})$"
  - $\Sigma \succeq 0 \iff v^\mathsf{T}\Sigma v \geq 0, \forall v \in \mathbb{R}^k$. $\lambda_i \geq 0, \forall i$
  - **Elements:** $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. $\Sigma_{ii} = \mathrm{Var}(X_i)$.
  - **Affine Map:** $A \in \mathbb{R}^{m \times k}$ and $b \in \mathbb{R}^m$ (constant):
    $\mathrm{Var}(A\mathbf{X} + b) = A\mathrm{Var}(\mathbf{X})A^\mathsf{T} = A\Sigma A^\mathsf{T}$

Transformations & Convolutions

- **1D Transform (monotonic)**: $Y = g(X) \implies$
  $f_Y(y) = f_X(g^{-1}(y))|\frac{d}{dy}g^{-1}(y)|$.
  $p_Y(y) = p_X(g^{-1}(y))$
- **1D Transform (general)**: $Y = g(X) \implies$
  $f_Y(y) = d_y F_Y(y) = d_y(\mathbb{P}(g(X) \leq y))$
- **2D Transform**: $f_{X,Y}(x, y) = f_{U,V}(u, v)|\det J|^{-1}$
  where Jacobian $J = \frac{\partial(x,y)}{\partial(u,v)}$.
- **Convolutions**: For $X \perp\!\!\!\perp Y$,
  $f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)dx$.

Advanced Conditioning

- **Conditional $\mathbb{E}$ and Var as a Random Variable**:
  $\mathbb{E}[X|Y] = g(Y)$, $\quad \mathrm{Var}(X|Y) = h(Y)$.
- **Iterated Expectation**: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$.
- **Total Var.**: $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X|Y)] + \mathrm{Var}(\mathbb{E}[X|Y])$.
- **Random Sums**: $Y = \sum_{i=1}^N X_i$, with N random:
  $\mathrm{Var}(Y) = \mathbb{E}[N]\mathrm{Var}(X) + \mathrm{Var}(N)(\mathbb{E}[X])^2$.

Bernoulli & Poisson Processes

Definitions & Arrival Counts

- **Bernoulli (Discrete)**: i.i.d. trials $X_i \in \{0, 1\}$.
  $\mathbb{P}(X_i = 1) = p$. $\mathbb{E}[X_i] = p, \mathrm{Var}(X_i) = p(1 - p)$.
- **Poisson (Continuous)**: Indep. increments. Small δ: $\mathbb{P}(1 \text{ arr}) = \lambda\delta + O(\delta^2), \mathbb{P}(0) = 1 - \lambda\delta + O(\delta^2), \mathbb{P}(> 1) = O(\delta^2)$.
- **Count Distributions**:
  - **Ber** (S_n in n slots): $S_n \sim \mathrm{Bin}(n, p) \implies$
    - $\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
    - $\mathbb{E}[S_n] = np, \mathrm{Var}(S_n) = np(1 - p)$.
  - **Poi** (N_τ in time τ): $N_\tau \sim \mathrm{Pois}(\lambda\tau) \implies$
    - $\mathbb{P}(N_\tau = k) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!} = P(k, \tau)$.
    - $\mathbb{E}[N_\tau] = \mathrm{Var}(N_\tau) = \lambda\tau$.
- **Approx.**: $\mathrm{Bin}(n, p) \xrightarrow{n,p \to \infty, 0} \mathrm{Pois}(\lambda = np)$.
- **Hack**: $\mathrm{Pois}(\lambda_1\tau_1) + \mathrm{Pois}(\lambda_2\tau_2) \sim \mathrm{Pois}(\lambda_1\tau_1 + \lambda_2\tau_2)$
  $N_\tau^A \perp\!\!\!\perp N_t^B \implies N_\tau^A + N_t^B \sim \mathrm{Poiss}(\lambda_A\tau + \lambda_B t)$

Interarrival (T_k) & k-th Arrival (Y_k) Times

- **Bernoulli (Discrete)**:
  $T_k \overset{\text{iid}}{\sim} \mathrm{Geom}(p)$ (Total trials).
  $Y_k \sim \mathrm{Pascal}(k, p)$ (Total trials).
- **Poisson (Continuous)**:
  $T_k \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda) \to \mathbb{E}[T_k] = 1/\lambda, \mathrm{Var}(T_k) = 1/\lambda^2$.
  $Y_k \sim \mathrm{Erlang}(k, \lambda) \to \mathbb{E}[Y_k] = k/\lambda, \mathrm{Var}(Y_k) = k/\lambda^2$.

Process Operations & Properties

- **Memorylessness**: Both restart fresh from any fixed/causally random time.
  For Poisson: if $T_1 > t \implies T_1 - t \sim \mathrm{Exp}(\lambda)$.
- **Merging Independent Processes**:
  - **Ber**: $\mathrm{Ber}(p) \oplus \mathrm{Ber}(q) \implies \mathrm{Ber}(p + q - pq)$.
    $\mathbb{P}(\text{arr is from process } 1) = \frac{p_1}{p_1 + p_2 - p_1 p_2}$.
    $\mathbb{P}(\text{arr is ONLY from process } 1) = \frac{p_1(1 - p_2)}{p_1 + p_2 - p_1 p_2}$.
  - **Poi**: $\mathrm{Pois}(\lambda_1) \oplus \mathrm{Pois}(\lambda_2) \implies \mathrm{Pois}(\lambda_1 + \lambda_2)$.
    $\mathbb{P}(\text{arr is from process } 1) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$.
- **Splitting Processes (with $\perp\!\!\!\perp$ coin bias q)**:
  - **Ber** $\implies \mathrm{Ber}(pq)$ and $\mathrm{Ber}(p(1 - q))$.
    Trap: Resulting Bern. streams are NOT indep.
  - **Poi** $\implies \mathrm{Pois}(\lambda q)$ and $\mathrm{Pois}(\lambda(1 - q))$.
    Ok: Resulting Poisson streams ARE indep.
- **Min/Max (Poisson Exponentials)**:
  For $X, Y, Z \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda) \implies \min \sim \mathrm{Exp}(3\lambda)$.
  $\mathbb{E}[\min] = \frac{1}{3\lambda}$ and $\mathbb{E}[\max] = \frac{1}{3\lambda} + \frac{1}{2\lambda} + \frac{1}{\lambda}$.
- **Ok:** Usefull trick true in general:
  $\min(X, Y) + \max(X, Y) = X + Y$

Random Incidence Paradox

- Arriving at random t^* to an ongoing Poisson process. Let U = last arrival, V = next arrival.
- Forward time $V - t^* \sim \mathrm{Exp}(\lambda)$
  Backward time $t^* - U \sim \mathrm{Exp}(\lambda)$
  $\implies \mathbb{E}[V - U] = 2/\lambda$.
- **Trap:** Expected interarrival time doubles ($2/\lambda$ instead of $1/\lambda$) because sampling larger intervals is proportionally more likely based on the sampling method.

Geometric Series

Arbitrary start index m (usually $= 0$). Requires $|r| < 1$.
$$\sum_{k=m}^{\infty} ar^k = \frac{ar^m}{1 - r}$$

Limit Theorems

Inequalities

- **Markov**: $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$ for $X \geq 0$.
- **Chebyshev**: $\mathbb{P}(|X - \mu| \geq c) \leq \sigma^2/c^2$.
- **Cauchy-Schwarz**: $|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$.
  . $\mathrm{Cov}(X, Y)^2 \leq \mathrm{Var}(X)\mathrm{Var}(Y)$.
- **Jensen**: $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$ for convex g. Reverse if concave.
- **Hoeffding**: If $X \in [a, b]$ a.s., then for all $\varepsilon > 0$:
  $$\mathbb{P}(|\overline{X}_n - \mu| \geq \varepsilon) \leq 2\exp\left(-\frac{2n\varepsilon^2}{(b - a)^2}\right)$$

Modes of Convergence

- **Almost Surely**: $\mathbb{P}(\lim_{n\to\infty} T_n = T) = 1$.
- **In Probability**: $\forall\varepsilon > 0, \mathbb{P}(|T_n - T| \geq \varepsilon) \xrightarrow{n\to\infty} 0$.
- **In Distribution**: $\mathbb{E}[f(T_n)] \xrightarrow{n\to\infty} \mathbb{E}[f(T)]$ for all continuous, bounded functions f.

$a.s. \implies$ in prob. $\implies$ in dist.
In dist. implies in prob. if the limit is a constant.
Operations: If $T_n \to T$ and $U_n \to U$ (a.s. or prob), then $T_n + U_n \to T + U$, $T_n U_n \to TU$, and $T_n/U_n \to T/U$ (if $U \neq 0$ a.s.).

Limit Theorems

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim}$ with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$.

- **WLLN**: $\overline{X_n} \xrightarrow{\mathbb{P}} \mu = \mathbb{E}[X_i]$.
- **SLLN**: $\overline{X_n} \xrightarrow{a.s.} \mu = \mathbb{E}[X_i]$.
- **CLT**: $\frac{\overline{X_n} - \mathbb{E}[X_i]}{\sqrt{\mathrm{Var}(\overline{X_n})}} = \sqrt{n}\frac{(\overline{X_n} - \mu)}{\sigma} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1)$ **Sums**: $S_n = \sum_1^n X_i \approx \mathcal{N}(n\mu, n\sigma^2)$.
- **Multivariate CLT**: $\sqrt{n}(\overline{X_n} - \mu) \xrightarrow{\mathcal{D}} \mathcal{N}_d(0, \Sigma)$

Slutsky, CMT & Delta Method

- **Slutsky's Theorem**: If $T_n \xrightarrow{\mathcal{D}} T$ and $U_n \xrightarrow{\mathbb{P}} u$ (constant): $T_n + U_n \xrightarrow{\mathcal{D}} T + u$, $T_n U_n \xrightarrow{\mathcal{D}} Tu$, $T_n/U_n \xrightarrow{\mathcal{D}} T/u$. $(u \neq 0.)$
- **Continuous Mapping (CMT)**: If f is continuous, $T_n \xrightarrow{a.s./\mathbb{P}/\mathcal{D}} T \implies f(T_n) \xrightarrow{a.s./\mathbb{P}/\mathcal{D}} f(T)$.
- **Delta Method ($\mathbb{R}$)**: For g contsly. difftbl. at θ:
  if $\sqrt{n}(Z_n - \theta) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$, then:
  $$\sqrt{n}(g(Z_n) - g(\theta)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, [g'(\theta)]^2\sigma^2)$$
- **Delta Method ($\mathbb{R}^d$)**: If $\sqrt{n}(T_n - \theta) \xrightarrow{\mathcal{D}} \mathcal{N}_d(0, \Sigma)$:
  $$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \nabla g(\theta)^\mathsf{T}\Sigma\nabla g(\theta))$$

Non trivial limits

If $c_1 = c_1(n)$, then the next limit is not trivial:
$$L = \lim_{n\to\infty} \Phi(2 * (c_1 - a)\sqrt{n}).$$
For example:
$$c_1 = a - \frac{b}{\sqrt{n}} \implies L = \Phi(2b).$$

Foundation of Inference

Statistical Model

Pair $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ where E is the sample space.

- **Parametric**: $\Theta \subseteq \mathbb{R}^d$.
- **Nonparametric**: Θ infinite dimensional.
- **Identifiability**: $\theta \neq \theta' \implies \mathbb{P}_\theta \neq \mathbb{P}_{\theta'}$.

Parameter Estimation

- **Consistency**: $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$ (weak) or $\xrightarrow{a.s.} \theta$ (strong).
- **Bias**: $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.
- **Quadratic Risk**: $R(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Var}(\hat{\theta}_n) + \text{Bias}(\hat{\theta}_n)^2$.
- **Jensen's**: Convex $f \implies \mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. Concave $g \implies \mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.

Confidence Intervals of level $1-\alpha$

Random interval $\mathcal{I}_\alpha$ not depending on θ such that: $\mathbb{P}_\theta(\mathcal{I}_\alpha \ni \theta) \geq 1 - \alpha$.

Distances and Divergences

- **Total Variation (TV)**: $\max_A |\mathbb{P}_\theta(A) - \mathbb{P}_{\theta'}(A)|$. Discrete: $\frac{1}{2}\sum |p_\theta(x) - p_{\theta'}(x)| \left(= \frac{1}{2}\int dx\right)$. (Metric: Symmetric, triangle inequality).
- **Kullback-Leibler (KL)**: $\mathbb{E}_\theta[\log(p_\theta(X)/p_{\theta'}(X))]$. Positive, definite, but **not** sym., no triangle ineq..

Methods of Estimation

Maximum Likelihood (MLE)

$\hat{\theta}_n^{\text{MLE}} = \text{argmax}_\theta\, \ell_n(\theta) = \text{argmax}_\theta \ln(L_n(X_{1:n};\theta))$.

- **Concavity**: ℓ_n is strictly concave $\iff \mathbb{H}\ell_n(\theta) \prec 0$.
- **Fisher Information**: In 1D: $I(\theta) = -\mathbb{E}[\ell_1''(\theta)]$. In general: $I(\theta) = -\mathbb{E}[\mathbb{H}\ell_1(\theta)]$.
- **Asymptotics**: Under regularity conditions (interior true param, identifiable, etc.):

$$\sqrt{n}(\hat{\theta}_n^{\text{MLE}} - \theta^*) \xrightarrow{\mathcal{D}} \mathcal{N}_d(0, I(\theta^*)^{-1})$$

Trap: The asymptotic Var is $I(\theta^*)^{-1}$, without n scaln.

Method of Moments (MM)

$m_k(\theta) = \mathbb{E}_\theta[X^k]$. $\hat{m}_k = \frac{1}{n}\sum X_i^k$.
$m_k(\hat{\theta}_n^{MM}) = \hat{m}_k$, for $k = 1, 2, \ldots, d$

- **Asymptotics**: $\sqrt{n}(\hat{\theta}_n^{MM} - \theta) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \Gamma(\theta))$ via Delta Method (uses Jacobian of M^{-1}).

MLE is generally more accurate, MM comput. easier.

M-Estimation

Estimate μ^* minimizing $Q(\mu) = \mathbb{E}[\rho(X_1, \mu)]$.

- $\rho = (x-\mu)^2 \implies \mu^*$ is the mean.
- $\rho = |x - \mu| \implies \mu^*$ is the median.
- $\rho = -\log L_1(x, \theta) \implies$ MLE.

Asymptotics: Let $J(\mu) = \mathbb{E}[\nabla^2 \rho(X, \mu)]$ and $K(\mu) = \text{Cov}(\nabla\rho(X,\mu))$. Then:
$\sqrt{n}(\hat{\mu}_n - \mu^*) \xrightarrow{\mathcal{D}} \mathcal{N}(0, J^{-1}KJ^{-1})$
(For log-likelihood MLE, $J = K = I$).

Hypothesis Testing

Testing Basics & Master Logic

1. State null / alternative: $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$.
2. Choose statistic $T_n(X_{1:n};\theta)$.
3. Determine distribution of T_n, under H_0: $T_n \sim ?Z?$.
4. Compute p-value or set rejection region R. Test $\psi = \mathbb{1}_{\{R\}}$.
5. Reject H_0 if p-value $\leq \alpha$ (or $\psi = 1$).
6. Always state assumptions.

- **Type I Error (α_ψ)**: Reject true H_0. Probability $\alpha_\psi(\lambda) = \mathbb{P}_\lambda(\psi = 1)$ for $\lambda \in \Theta_0$.
- **Type II Error (β_ψ)**: Fail to reject false H_0. Probability $\beta_\psi(\lambda) = \mathbb{P}_\lambda(\psi = 0)$ for $\lambda \in \Theta_1$.
- **Power (π_ψ)**: $1 - \beta(\theta)$ for $\theta \in \Theta_1$. Often $\inf_{\theta\in\Theta_1}(1 - \beta_\psi(\theta))$.
- **Test Level/Size**: $\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(\psi = 1) \leq \alpha$. Controls worst-case Type I error probability.
- **Asympt. Level**: $\lim_{n\to\infty} \sup_{\theta\in\Theta_0} \mathbb{P}_\theta(\psi_n = 1) \leq \alpha$
- **p-value**: Under H_0, smallest level α at which H_0 is rejected, or probability of result at least as extreme as observed.
- **CI / Test Duality**: For many regular problems: Reject $H_0 : \theta = \theta_0$ at level $\alpha \iff \theta_0 \notin (1-\alpha)$ CI.

Trap: Failing to reject H_0 does *not* prove H_0. Statistical significance $\neq$ practical significance.

Power Reminders

Power increases with: Larger sample size n, larger effect size, smaller variance / noise, larger α (but Type I error also increases).

Quantiles

- **Quantile of order $1-\alpha$** is the number q_α defined by:
  - Lower Tail: $\mathbb{P}(X \leq q_\alpha) = 1 - \alpha$
  - Upper Tail: $\mathbb{P}(X > q_\alpha) = \alpha$. (Top α.)
  - CDF Relation: $F(q_\alpha) = 1 - \alpha \implies q_\alpha = F^{-1}(1-\alpha)$
- **Examples**: $\mathbb{P}(X \leq q_{0.05}) = 0.95$, $\mathbb{P}(X \leq q_{0.7}) = 0.3$.
- **Quantiles of Standard Normal ($X \sim \mathcal{N}(0,1)$)**
  - Sym. $q_{1-\alpha} = -q_\alpha$. $\mathbb{P}(|X| \leq q_{\alpha/2}) = 1 - \alpha$.
  - $\mathbb{P}(|X| > q_{\alpha/2}) = \alpha = 2(1 - \Phi(q_{\alpha/2}))$.
  - Common Values: $q_{0.025} = 1.96$, $q_{0.05} = 1.645$

Exam Rescue Strip

1. Identify parameter of interest.
2. Identify data structure: 1-sample / 2-sample / paired / count / proportion.
3. Check assumptions: normal? indep? known σ?
4. Choose exact vs asymptotic method.
5. Write statistic, null distribution, rejection rule / p-value.
6. Interpret in words, including assumptions.

Trap: For two-sided tests, use both tails / absolute value logic. For one-sided tests, align rejection direction with H_1. Check whether variance is known, unknown, equal, or unequal before choosing z / pooled t / Welch.

One-Sample Tests

- **Wald Test (1D)**: (For p-vals $\mathbb{P} = \mathbb{P}_{\theta_0}$.)

	$H_0 : \theta = \theta_0$, $H_1 : \theta \neq \theta_0$	$H_0 : \theta \leq \theta_0$, $H_1 : \theta > \theta_0$	$H_0 : \theta \geq \theta_0$, $H_1 : \theta < \theta_0$
Wald test ψ	$\mathbb{1}\{\lvert W\rvert > q_{\alpha/2}\}$	$\mathbb{1}\{W > q_\alpha\}$	$\mathbb{1}\{W < -q_\alpha\}$
p-value	$\mathbb{P}(\lvert W\rvert > \lvert W^{\text{obs}}\rvert)$	$\mathbb{P}(W > W^{\text{obs}})$	$\mathbb{P}(W < W^{\text{obs}})$
asymp. p-val	$\mathbb{P}(\lvert Z\rvert > \lvert W^{\text{obs}}\rvert)$	$\mathbb{P}(Z > W^{\text{obs}})$	$\mathbb{P}(Z < W^{\text{obs}})$

- $\sqrt{n}$ **scaling**: $W_n \xrightarrow{\mathcal{D}} \mathcal{N}(0,1)$.

$$W_n = \frac{(\hat{\theta}_n - \theta_0)}{\sqrt{\text{Var}(\hat{\theta})}} = \sqrt{nI(\hat{\theta}_n)}(\hat{\theta}_n - \theta_0) \xrightarrow[H_0]{\mathcal{D}} \mathcal{N}(0,1)$$

- n **scaling**: $T_n \xrightarrow{\mathcal{D}} \chi_1^2$. Test $\psi = \mathbb{1}\{T_n > q_\alpha^{(\chi_1^2)}\}$:

$$T_n = nI(\hat{\theta}_n)(\hat{\theta}_n - \theta_0)^2 \xrightarrow{\mathcal{D}} \chi_1^2 \text{ under } H_0$$

Trap: We dont need to use MLE, but we need consistent $(\sqrt{n})\sqrt{\text{Var}(\hat{\theta})}$, and asymptotically normal W_n.

- **Student's T-test for μ**: $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2(\text{unknown}))$.

	$H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$	$H_0 : \mu \leq \mu_0$, $H_1 : \mu > \mu_0$	$H_0 : \mu \geq \mu_0$, $H_1 : \mu < \mu_0$
T-Test ψ	$\mathbb{1}\{\lvert T\rvert > q_{\alpha/2}^{t_{n-1}}\}$	$\mathbb{1}\{T > q_\alpha^{t_{n-1}}\}$	$\mathbb{1}\{T < -q_\alpha^{t_{n-1}}\}$

$$T_n = \sqrt{n}\frac{(\overline{X}_n - \mu_0)}{\sqrt{\tilde{S}_n^2}} \sim t_{n-1} \text{ under } H_0$$

Trap: $\tilde{S}_n^2$ UNBIASED! $\tilde{S}_n^2 = \frac{n}{n-1}S_{pop}^2$

- **Test for variance**: $X_{in} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2(\text{unknown}))$.

$$T_n = \frac{(n-1)\tilde{S}_n^2}{\sigma_0^2} \sim \chi_{n-1}^2 \text{ under } H_0$$

- ***z*-test for mean**: If $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ or large n and σ known: $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$ under H_0
- **Proportion test**: If $X \sim \text{Bin}(n,p)$ and n large: $Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \approx \mathcal{N}(0,1)$

Two-Sample Tests

- **2-S. T-test**: $X_{in} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_x, \sigma_x^2), Y_{im} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_y, \sigma_y^2)$. $H_0 : \mu_x - \mu_y \leq 0$, $H_1 : \mu_x - \mu_y > 0$.

$$T = \frac{(\overline{X_n} - \overline{Y_m}) - (\mu_x - \mu_y)}{\sqrt{S_X^2/n_X + S_Y^2/n_Y}} \sim t_N$$

Approx. dof (Welch–Satterthwaite):

$$N \approx \frac{(S_X^2/n + S_Y^2/m)^2}{\frac{\left(S_X^2/n\right)^2}{n-1} + \frac{\left(S_Y^2/m\right)^2}{m-1}} \gtrapprox \min(n, m)$$

Fisher's Exact Test: H_0 : Variables are $\perp\!\!\!\perp$. Assumes H_0 and the marginal totals of the table are fixed.

	B_1	B_2	Total
A_1	a	b	$a+b$
A_2	c	d	$c+d$
Total	$a+c$	$b+d$	n

$$T = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{n}{a+c}}$$

Likelihood-Ratio Test (LRT)

$H_0 : \theta \in \Theta_0$. (Fix $d - r$ params.)
With MLEs $\hat{\theta}_n^{\text{MLE}}$ (all) and $\hat{\theta}_n^{\text{MLE}_C}$ (constrained):

$$T_n = -2\ln\left(\frac{\sup_{\Theta_0} L_n(\theta)}{\sup_\Theta L_n(\theta)}\right) = -2\ln\left(\frac{L_n(\hat{\theta}_n^{\text{MLE}_C})}{L_n(\hat{\theta}_n^{\text{MLE}})}\right)$$

$T_n = 2\left(\ell_n(\hat{\theta}_n^{\text{MLE}}) - \ell_n(\hat{\theta}_n^{\text{MLE}_C})\right) \xrightarrow{\mathcal{D}} \chi_{d-r}^2$
$\psi = \mathbb{1}\{T_n > q_\alpha(\chi_{d-r}^2)\}$, $\mathbf{d} - \mathbf{r} = \dim(\Theta) - \dim(\Theta_0)$.
Trap: We fix the complement of Θ_0 (r params.)

Nonparametric Tests & GOF

Goodness of Fit (Discrete / χ^2)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} P$ on the support $\{0, \ldots, K\}$.

- $|\text{support}| = K + 1$.
- Cell counts $N_j := \sum_{i=1}^n \mathbb{1}\{X_i = j\}$. $n = \sum_{j=0}^K N_j$.
- **Hypotheses & Test Statistics**:
  - **Parametric family:** $H_0 :\ P \in \{\mathbb{P}_\theta\}_{\theta\in\Theta\subseteq\mathbb{R}^d}$ vs $H_1 :\ P \notin \{\mathbb{P}_\theta\}_{\theta\in\Theta\subseteq\mathbb{R}^d}$.
    - $*$Fit params: $\hat{\theta}_0 = \hat{\theta}_{H_0}^{\text{MLE}}$. Let $f_{\hat{\theta}}(j) := \mathbb{P}_{\hat{\theta}}(X = j)$.
    - $$*T_n = n\sum_{j=0}^K \frac{\left(N_j/n - f_{\hat{\theta}}(j)\right)^2}{f_{\hat{\theta}}(j)} \xrightarrow[H_0]{\mathcal{D}} \chi_{K-d}^2$$

    Trap: For $\hat{\theta}$ use counts in bins, not the original X_i's.
  - **Fixed distribution:** $H_0 : \mathbf{p} = \mathbf{p}^0 \in \Delta_K$, $\hat{p}_j = N_j/n$.
    - $$*T_n = n\sum_{j=0}^K \frac{\left(\hat{p}_j - p_j^0\right)^2}{p_j^0} \xrightarrow[H_0]{\mathcal{D}} \chi_K^2$$
- **Equivalent Form (Observed vs Expected)**:
  - Observed $O_j := N_j$, Expected $E_j := np_j$ (using p_j^0 or $f_{\hat{\theta}}(j)$ depending on H_0).
  - $$T_n = \sum_{j=0}^K \frac{(O_j - E_j)^2}{E_j} \xrightarrow[H_0]{\mathcal{D}} \chi_{K-d}^2$$
- **Degrees of Freedom: K – d**:
  - Support size is $K + 1$.
  - Subtract 1 DoF due to sum constraint $\sum N_j = n$.
  - Subtract d DoF for estimating d parameters. (**Note:** $d = 0$ for a Fixed Distribution).
  - Total DoF $= (K+1) - 1 - d = K - d$.

Trap: If p_j's are fully known (i.e., no parameters are fitted), use: *DoF* $= (K+1) - 1 = K$, not $K - d$. More generally, if the support has m cells, use: *DoF* $= m - 1 - d$. Always count the number of fitted parameters actually estimated from the data.

Goodness of Fit (Continuous / CDF)

Empirical CDF: $F_n(t) = \frac{1}{n}\sum \mathbb{1}\{X_i \leq t\}$.
Glivenko-Cantelli: $\sup_t |F_n(t) - F(t)| \xrightarrow{a.s.} 0$.
Donsker's: $\sqrt{n}\sup_t |F_n(t) - F(t)| \xrightarrow{\mathcal{D}} \sup_t |B(t)|$.

Kolmogorov-Smirnov (KS) Test: **Pivotal** for any sample size n, provided: The null distribution is continuous. The null distribution is fully specified.

$$\text{Test: } \psi = \mathbb{1}\{T_n > q_\alpha^{\text{KS}}\}$$

$$T_n = \sup_{t\in\mathbb{R}} \sqrt{n}|F_n(t) - F^0(t)| \sim$$

Let $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$ be the reordered sample. The expression for T_n reduces to:

$$T_n = \sqrt{n}\max_{1\leq i\leq n}\max\left(D_i^-, D_i^+\right)$$

$$D_i^- = \left|\frac{i-1}{n} - F^0\left(X_{(i)}\right)\right|, D_i^+ = \left|\frac{i}{n} - F^0\left(X_{(i)}\right)\right|$$

Kolmogorov-Lilliefors (KL) Test: Tests Gaussianity where params $(\hat{\mu}, \hat{\sigma}^2)$ are estimated. *Donsker* is invalid; uses different quantiles.

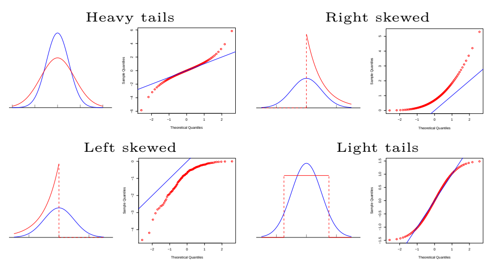
$$\text{Test: } \psi = \mathbb{1}\{T_n > q_\alpha^{\text{KL}}\}$$

$$T_n = \sup_{t\in\mathbb{R}}\left|F_n(t) - \Phi_{\hat{\mu},\hat{\sigma}^2}(t)\right|$$

where $\hat{\mu} = \overline{X}_n$, $\hat{\sigma}^2 = S_n^2$.

Q-Q Plots

Plot $(F^{-1}(i/n), F_n^{-1}(i/n))$ against $y = x$. Visually checks if distributions match.



Horiz. = Theoretical quantiles. Vert. = Sample quant.

Multiple Testing

FWER: Family-Wise Error Rate

Bonferroni Method (FWER)

Controls **FWER**: ($\leq \alpha$.)

- **Rule**: Reject the i-th test $\iff P_i \leq \alpha/N$.
- **Proof**: $\text{FWER} = \mathbb{P}_{\mu_i=0}\left(\bigcup_{i=1}^{N}\{P_i < \frac{\alpha}{N}\}\right)$
  $\text{FWER} \leq \sum_{i=1}^{N} \mathbb{P}_{\mu_i=0}(|T_i| > q_{\frac{\alpha}{2N}}) = N \cdot \frac{\alpha}{N} = \alpha$.

Trap: Often way too conservative, leading to low power (no discoveries).

Holm-Bonferroni Method (FWER)

Controls **FWER**: ($\leq \alpha$.)

- **Ordering**: Order p-values: $P_{(1)} \leq P_{(2)} \leq \cdots \leq P_{(N)}$.
- **Rule**: Reject $H_{(i)}$ if $P_{(j)} \leq \frac{\alpha}{N-j+1}$ for all $j \leq i$. (Step-down: Stop at the first i where $P_{(i)} > \frac{\alpha}{N-i+1}$ and fail to reject all subsequent hypotheses).

Ok: Uniformly more powerful than the standard Bonferroni method.

FDR: False Discovery Rate

Benjamini-Hochberg Method (FDR)

Controls **FDR**: ($\leq \alpha$.)

- **Ordering**: Order p-values: $P_{(1)} \leq P_{(2)} \leq \cdots \leq P_{(N)}$.
- **Rule**: Let $i_{\max} := \max\left\{i : P_{(i)} < i \cdot \frac{\alpha}{N}\right\}$. Reject all tests (i) such that $i \leq i_{\max}$.

Ok: Intuitively rejects tests with the smallest p-values and is less conservative than (Holm-)Bonferroni.

Bayesian Statistics & Inference

Basics, Priors and Posteriors

- **Universal Bayes**:

$$\Pr(\theta|x) = \frac{\Pr(\theta)\Pr(x|\theta)}{\Pr(x)}$$

Notation: $Pr = f(=p)$, f if cont. (p if disc.).
$\Pr(x) = \sum_{\Theta} \Pr(\theta')\Pr(x|\theta') \left(= \int_{\Theta} d\theta'\right)$

- **Bayesian Setup**: Prior $\pi(\theta)$. (or $p_\Theta(\theta)$, or $f_\Theta(\theta)$). Likelihood $L_n(X_{1:n}|\theta)$. (or $p_{X|\Theta}(x|\theta)$, $f_{X|\Theta}(x|\theta)$).
- **Bayes' Formula / Posterior**:
  $\pi(\theta|X_{1:n}) = \pi_{\Theta|X_{1:n}}(\theta|x_{1:n})$

$$\pi(\theta|X_{1:n}) = \frac{\pi(\theta)L_n(X_{1:n}|\theta)}{L_n(X_{1:n})} \propto \pi(\theta)L_n(X_{1:n}|\theta).$$

Marginal $L_n(X_{1:n}) = \int_{\Theta} \pi(\theta)L_n(X_{1:n}|\theta)d\theta$.

Trap: Deterministic Priors dont change given any data.

Bayesian Estimators

- **Posterior Mean/LMS**: $\hat{\theta}^{(\pi)} = \hat{\theta}_{\text{LMS}} = \mathbb{E}[\Theta|X_{1:n}]$
  $\hat{\theta}^{\pi} = \int_{\Theta} \theta\pi(\theta|X_{1:n})d\theta = \text{argmin}_{\hat{\theta}}\, \mathbb{E}[(\Theta - \hat{\theta})^2|X_{1:n}]$.
- **MAP**: $\hat{\theta}_{\text{MAP}} = \text{argmax}_{\theta}\, \pi(\theta|X_{1:n})$.
- **LLMS**:
  $\hat{\Theta}_{LLMS} = \mathbb{E}[\Theta] + \frac{\text{Cov}(\Theta,X)}{\text{Var}(X)}(X - \mathbb{E}[X])$.
- **Confidence Region of level α**: deterministic subset $\mathcal{R}$ such that $\mathbb{P}(\theta \in \mathcal{R}|X_{1:n}) = 1 - \alpha$. Depends on prior choice.

Conjugate Priors

- **Bernoulli-Beta**: if $p \sim \text{Beta}(a, b)$, $X_i|p \sim \text{Bern}(p)$.
  $\implies p|X_{1:n} \sim \text{Beta}(a + \sum X_i, b + n - \sum X_i)$.
  Prior pdf: $\pi(p) \propto p^{a-1}(1-p)^{b-1}$.
  Likelihood: $L_n \propto p^{\sum X_i}(1-p)^{n-\sum X_i}$.
  $\pi(p|X_{1:n}) \propto p^{(a+\sum X_i)-1}(1-p)^{(b+n-\sum X_i)-1}$.
- **Poiss-Gamma**: $\Lambda \sim \text{Gam}(\alpha, \lambda)$, $X_i|\Lambda \sim \text{Poiss}(\Lambda)$.
  $\implies \Lambda|X_{1:n} \sim \text{Gamma}(\alpha + \sum X_i, \lambda + n)$
- **Normal-Normal**: Normal mean + Normal prior.
  $\implies$ posterior Normal.

Non-informative & Jeffreys Priors

- **Improper prior**: $\int \pi(\theta)d\theta = \infty$ (e.g., $\propto 1$).
  Ok: Can still yield a proper posterior.
- **Jeffreys Prior**: $\pi_J(\theta) \propto \sqrt{\det I(\theta)}$. if $\eta = \phi(\theta)$:

$$\pi_J(\phi^{-1}(\eta))\left|\frac{1}{\phi'(\phi^{-1}(\eta))}\right| = \tilde{\pi}_J(\eta) \propto \sqrt{\det \tilde{I}(\eta)}.$$

i.e., satisfies **reparameterization invariance**.
$-X \sim \text{Bern}(p) \implies \pi_J(p) \propto p^{-1/2}(1-p)^{-1/2}$.
$-X \sim \mathcal{N}(\mu, \sigma^2(\text{known})) \implies \pi_J(\mu) \propto 1$.

Linear Regression

Modeling Assumptions

Describe $\mathbb{P}$ partially via regression function: $f(x) = \mathbb{E}[Y|X = x]$. Theoretical min of $\mathbb{E}[(Y - a - bX)^2]$ gives: $b^* = \text{Cov}(X, Y)/\text{Var}(X)$, $a^* = \mathbb{E}[Y] - b^*\mathbb{E}[X]$.
Noise: $\varepsilon = Y - (a^* + b^*X)$. $\mathbb{E}[\varepsilon] = 0$, $\text{Cov}(X, \varepsilon) = 0$.
1D LSE:

$$\hat{b} = \frac{\overline{XY} - \bar{X}\bar{Y}}{\overline{X^2} - \bar{X}^2}, \quad \hat{a} = \bar{Y} - \hat{b}\bar{X}$$

Multivariate LSE

Matrix Form: $\mathbf{Y} = \mathbb{X}\beta^* + \boldsymbol{\varepsilon}$. Assuming $\text{rank}(\mathbb{X}) = p$:

$$\hat{\boldsymbol{\beta}} = \underset{\beta}{\text{argmin}}\, \|\mathbf{Y} - \mathbb{X}\beta\|_2^2 = (\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top\mathbf{Y}$$

Geometric: $\mathbb{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{Y}$ onto the column space of $\mathbb{X}$.
Projection matrix: $P = \mathbb{X}(\mathbb{X}^\top\mathbb{X})^{-1}\mathbb{X}^\top$.

Statistical Inference for LSE

Assume $\mathbb{X}$ deterministic, homoscedastic ($\varepsilon_i \overset{\text{iid}}{\sim}$), Gaussian noise $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$.

- **Distribution**: $\hat{\beta} \sim \mathcal{N}_p(\beta^*, \sigma^2(\mathbb{X}^\top\mathbb{X})^{-1})$.
- **Risk**: $\mathbb{E}[\left\|\hat{\beta} - \beta^*\right\|_2^2] = \sigma^2\,\text{tr}((\mathbb{X}^\top\mathbb{X})^{-1})$.
- **Prediction Error**: $\mathbb{E}[\left\|\mathbf{Y} - \mathbb{X}\hat{\beta}\right\|_2^2] = \sigma^2(n - p)$.
- **Unbiased Variance**: $\hat{\sigma}^2_{unb} = \frac{1}{n-p}\left\|\mathbf{Y} - \mathbb{X}\hat{\beta}\right\|_2^2$ (with residuals $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbb{X}\hat{\beta}$).

Hypothesis Testing Framework & Contrasts

Evaluate linear combinations via a contrast vector $u \in \mathbb{R}^p$.

- **Example:** $u = [1, -1, 0, \ldots, 0]^\top$ isolates β_1 and β_2 to test if $\beta_1 \leq \beta_2$.
- **Hypotheses:** $H_0 : u^\top\beta^* \leq 0$ vs $H_1 : u^\top\beta^* > 0$.

1. Pivotal Quantity Construction The test statistic is built from independent Normal and χ^2 components:

- **Numerator (Normal)**: The normalized estimator:

$$Z = \frac{u^\top\hat{\beta} - u^\top\beta^*}{\sigma\sqrt{u^\top(\mathbb{X}^\top\mathbb{X})^{-1}u}} \sim \mathcal{N}(0, 1)$$

- **Denominator (Chi-Squared)**: The variance ratio:

$$V = \frac{\hat{\sigma}^2_{unb}}{\sigma^2} \sim \frac{\chi^2_{n-p}}{n-p}$$

- **Student's t Pivot**: Because $\hat{\sigma}^2_{unb} \perp\!\!\!\perp \hat{\beta}$, the unknown true variance σ^2 cancels out when taking $Z/\sqrt{V}$:

$$\frac{u^\top\hat{\beta} - u^\top\beta^*}{\hat{\sigma}_{unb}\sqrt{u^\top(\mathbb{X}^\top\mathbb{X})^{-1}u}} \sim t_{n-p}$$

2. Execution Recipe

1. **Fit Model**: Calculate $\hat{\beta}$, residuals $\hat{\boldsymbol{\varepsilon}}$, and $\hat{\sigma}^2_{unb}$.
2. **Standard Error**: Compute the specific error for the contrast:

$$\text{se}(u^\top\hat{\beta}) = \hat{\sigma}_{unb}\sqrt{u^\top(\mathbb{X}^\top\mathbb{X})^{-1}u}$$

3. **Test Statistic (T_n)**: Evaluate under the strict boundary condition of the null hypothesis ($u^\top\beta^* = 0$):

$$T_n = \frac{u^\top\hat{\beta}}{\text{se}(u^\top\hat{\beta})}$$

4. **Decision Rule**: Find the critical threshold $s = q_{1-\alpha}(t_{n-p})$. Reject H_0 if $T_n > s$.

Trap: Type I Error Control: If the true parameter is strictly interior ($u^\top\beta^* < 0$), the actual probability of falsely rejecting H_0 is strictly bounded below α. The threshold s guarantees exact α-level control right at the boundary.

Variable Significance Tests

- **Single Variable**: $H_0 : \beta_j = 0$. Let $\gamma_j = [(\mathbb{X}^\top\mathbb{X})^{-1}]_{jj}$. The test statistic is:

$$T_n^{(j)} = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2_{unb}\gamma_j}} \sim t_{n-p}$$

Reject if $|T_n^{(j)}| > q_{\alpha/2}(t_{n-p})$.

- **Bonferroni Correction**: Test a group of variables S. Reject if any $|T_n^{(j)}| > q_{\alpha/(2k)}$ where $k = |S|$.

Data Analysis

Causality

The fundamental problem of causal inference: We can never observe the **counterfactual**. It is impossible to simultaneously observe both the treated and untreated outcomes for the same subject.

Randomized Control Trials (RCTs)

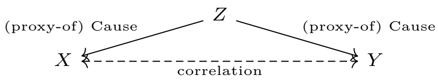
Important: **Consent** is important, correctly **informing** the participants.
We can try to identify **confounder** variables before randomizing, **stratify**, and **randomize** within each group. Then aggregate results.

- **Randomized**: Causality problem, neutralizing both known and unknown confounding variables.
- **Controlled**: Involves at least one control group (comparative counterfactual).
- **Trial**: A study design where researchers administer a treatment and track outcomes moving forward.
- **Double-Blind**: Participants & researchers don't know who is in which group till end. (placebo effect, confirmation or observer bias).

Confounders If we are trying to identify the relation between X and Y. A confounder is a variable Z that affects both variables or is affected by an underlying cause U that affects both variables. Making X and Y correlated without having a causal relation between them.

Trap: Assuming a reversed causality because of a confounder. Example below:

Source of Confounding



p-values: If we run a poorly designed experiment to get data to regress Y on $\mathbf{X}$, even with a lot of data, getting small p-value for some components of $\mathbf{X}$ indicated a valid correlation, **not causation**. Could be used as a *first approach* to identify possible causal **theories/mechanism**, necessary for **scientific** claims.

- **Metric of Incompatibility:** A p-value indicates the degree of incompatibility between observed data and a specified statistical model constructed under a set of assumptions, along with a "null hypothesis".
- **Misinterpretation of Probability:** It does *not* measure the probability that a studied hypothesis is true or that the data were produced by random chance alone.
- **Threshold Limitations:** Scientific, business, and policy decisions should not rely solely on whether a p-value crosses a specific threshold (e.g., $p < 0.05$). P-values supplement, but do not replace, scientific reasoning.
- **Integrity through Transparency:** Proper inference requires full reporting. Selective reporting (*p-hacking*) renders conclusions unreliable, whereas transparency regarding all tests conducted minimizes bias.
- **Independence from Magnitude:** Statistical significance does not measure effect size or the importance of a result. Separate estimation methods, such as confidence intervals or Bayesian approaches, should be used to quantify uncertainty and effect magnitude.

Notes: Individual $\neq$ Aggregate. Is the variable measured a good proxy for variable of interest?

W. Most Pub. Research Find. is False

- ***R* (Pre-study Odds):** Ratio of true relationships to no relationships.
- ***u* (Bias):** Prop. of analyses reported as findings due to manipulation/selective reporting.
- ***n* (Teams):** Number of independent research teams testing the same hypothesis.
- **Positive Predictive Value (PPV)** The post-study probability that a statistically significant finding is actually true: $PPV = \frac{\text{True Positives}}{\text{Total Positive Findings}}$

The Six Corollaries Research findings are *less* likely to be true when:

1. **Small Studies:** $\downarrow$ sample sizes yield $\downarrow$ statistical power $(1 - \beta)$.
2. **Small Effect Sizes:** Postulated effects are barely distinguishable from noise.
3. **Massive Testing:** High-throughput testing of thousands of hypotheses shrinks R.
4. **High Flexibility:** Broad freedom in study design, definitions, and analytical modes.
5. **Interest & Prejudice:** Financial/personal conflicts artificially inflate bias (u).
6. **"Hot" Fields:** $\uparrow$ independent teams (n) competing for impressive results $\to\uparrow$ false positives.

Assumption: We consider an i.i.d. sample $X_1, \ldots, X_n$ from the specified distribution, where $\overline{X_n} = \frac{1}{n}\sum_{i=1}^n X_i$ denotes the sample mean.

**Note: For the Binomial and Hypergeometric distributions, the traditional parameter n (number of trials/draws) has been renamed to m to avoid confusion with the sample size n.*

Distribution	**PMF/PDF & support**	$\mathbb{E}[X]$	$\mathrm{Var}(X)$	**MLE** $\hat{\theta}$	$\mathcal{I}(\theta)$	**Asymp. Var.** V ($\mathrm{Var}(\hat{\theta}) \approx V/n$)
Bernoulli $\mathrm{Bern}(p)$ $p \in [0,1]$ *Indicator of success.*	$\mathbb{P}(X=k) = p^k(1-p)^{1-k}$ $k \in \{0,1\}$	p	$p(1-p)$	$\overline{X_n}$	$\dfrac{1}{p(1-p)}$	$p(1-p)$
Binomial $\mathrm{Bin}(m,p)$ $m \in \mathbb{Z}^+,\ p \in [0,1]$ *Successes in m indep. trials.*	$\mathbb{P}(X=k) = \binom{m}{k} p^k (1-p)^{m-k}$ $k \in \{0,1,\ldots,m\}$	mp	$mp(1-p)$	$\dfrac{\overline{X_n}}{m}$ (if m known)	$\dfrac{m}{p(1-p)}$	$\dfrac{p(1-p)}{m}$
Geometric (Total Failures) $\mathrm{Geom}(p)$ $p \in [0,1]$ *Failures before 1st success.*	$\mathbb{P}(X=k) = (1-p)^k p$ $k \in \{0,1,2,\ldots\}$	$\dfrac{1-p}{p}$	$\dfrac{1-p}{p^2}$	$\dfrac{1}{1+\overline{X_n}}$	$\dfrac{1}{p^2(1-p)}$	$p^2(1-p)$
Geometric (Total Trials) $\mathrm{Geom}(p)$ $p \in [0,1]$ *Trials until 1st success.*	$\mathbb{P}(X=t) = (1-p)^{t-1} p$ $t \in \{1,2,\ldots\}$	$\dfrac{1}{p}$	$\dfrac{1-p}{p^2}$	$\dfrac{1}{\overline{X_n}}$	$\dfrac{1}{p^2(1-p)}$	$p^2(1-p)$
Neg.Binomial / Pascal (Total Failures r) $\mathrm{NBin}(k,p)$ (k arrivals) $k \in \mathbb{Z}^+,\ p \in [0,1]$ *Failures before kth success.*	$\mathbb{P}(X=r) = \binom{r+k-1}{k-1} p^k (1-p)^r$ $r \in \{0,1,2,\ldots\}$	$\dfrac{k(1-p)}{p}$	$\dfrac{k(1-p)}{p^2}$	$\dfrac{k}{k+\overline{X_n}}$ (if k known)	$\dfrac{k}{p^2(1-p)}$	$\dfrac{p^2(1-p)}{k}$
Neg.Binomial / Pascal (Total Trials t) $\mathrm{Pascal}(k,p)$ (k arrivals) $k \in \mathbb{Z}^+,\ p \in [0,1]$ *Trials until kth success.*	$\mathbb{P}(X=t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}$ $t \in \{k,k+1,\ldots\}$	$\dfrac{k}{p}$	$\dfrac{k(1-p)}{p^2}$	$\dfrac{k}{\overline{X_n}}$ (if k known)	$\dfrac{k}{p^2(1-p)}$	$\dfrac{p^2(1-p)}{k}$
Poisson $\mathrm{Pois}(\lambda)$ $\lambda > 0$ *Rare events.*	$\mathbb{P}(X=k) = e^{-\lambda}\dfrac{\lambda^k}{k!}$ $k \in \{0,1,2,\ldots\}$	λ	λ	$\overline{X_n}$	$\dfrac{1}{\lambda}$	λ
Uniform $\mathrm{Unif}(a,b)$ $a < b$	$f(x) = \dfrac{1}{b-a}$ $x \in [a,b]$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$	$\hat{a} = X_{(1)}$ (biased) $\hat{b} = X_{(n)}$ (biased)	N/A	$\mathcal{O}(1/n^2)$
Normal $\mathcal{N}(\mu,\sigma^2)$ $\mu \in \mathbb{R},\ \sigma^2 > 0$ *Standardizes to $\mathcal{N}(0,1)$*	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$ $x \in \mathbb{R}$ $\mathbb{E}[(X-\mu)^p] = \sigma^p(p-1)$ for p even ($=0$ ow.)	μ	σ^2	$\hat{\mu} = \overline{X_n}$ $\hat{\sigma}^2 = S^2_{\text{pop}}$ (biased)	$\mathrm{diag}\left(\dfrac{1}{\sigma^2}, \dfrac{1}{2\sigma^4}\right)$	$\mathrm{diag}(\sigma^2, 2\sigma^4)$
Multiv. Normal $\mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ $\boldsymbol{\mu} \in \mathbb{R}^d,\ \boldsymbol{\Sigma} \succ 0$	$f(\mathbf{x}) = \dfrac{1}{\sqrt{(2\pi)^d \lvert\boldsymbol{\Sigma}\rvert}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\intercal \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$ $\mathbf{x} \in \mathbb{R}^d$	$\boldsymbol{\mu}$	$\boldsymbol{\Sigma}$	$\hat{\boldsymbol{\mu}} = \overline{\mathbf{X}_n}$ $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{X}_i - \hat{\boldsymbol{\mu}})$ $\times(\mathbf{X}_i - \hat{\boldsymbol{\mu}})^\intercal$	Block matrix involving $\boldsymbol{\Sigma}^{-1}$	$\mathcal{I}(\boldsymbol{\mu},\boldsymbol{\Sigma})^{-1}$
Exponential $\mathrm{Expo}(\lambda)$ $\lambda > 0$ *Memoryless:* $\mathbb{P}(X_{>s+t} \mid X_{>s}) = \mathbb{P}(X_{>t})$	$f(x) = \lambda e^{-\lambda x}$ $x \in [0,\infty)$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$	$\dfrac{1}{\overline{X_n}}$	$\dfrac{1}{\lambda^2}$	λ^2
Gamma (Rate) / Erlang $\mathrm{Gamma}(a,\lambda)$ $a > 0,\ \lambda > 0$ $\mathrm{Erlang}(a = k \in \mathbb{Z}^+)$ (k arrivals, $\Gamma(k) = (k-1)!$) *Sum of a iid $\mathrm{Expo}(\lambda)$.*	$f(x) = \dfrac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}$ $x \in (0,\infty)$	$\dfrac{a}{\lambda}$	$\dfrac{a}{\lambda^2}$	$\hat{a}$: No closed form $\hat{\lambda} = \hat{a}/\overline{X_n}$	$\begin{pmatrix} \psi'(a) & -\frac{1}{\lambda} \\ -\frac{1}{\lambda} & \frac{a}{\lambda^2} \end{pmatrix}$	$\mathcal{I}(a,\lambda)^{-1}$
Gamma (Scale) $\mathrm{Gamma}(k,\theta)$ $k > 0,\ \theta > 0$	$f(x) = \dfrac{1}{\theta^k \Gamma(k)} x^{k-1} e^{-x/\theta}$ $x \in (0,\infty)$	$k\theta$	$k\theta^2$	$\hat{k}$: No closed form $\hat{\theta} = \overline{X_n}/\hat{k}$	$\begin{pmatrix} \psi'(k) & \frac{1}{\theta} \\ \frac{1}{\theta} & \frac{k}{\theta^2} \end{pmatrix}$	$\mathcal{I}(k,\theta)^{-1}$
Inverse Gamma $\mathrm{InvGamma}(\alpha,\beta)$ $\alpha > 0,\ \beta > 0$	$f(x) = \dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$ $x \in (0,\infty)$	$\dfrac{\beta}{\alpha-1}$	$\dfrac{\beta^2}{(\alpha-1)^2(\alpha-2)}$ ($\alpha > 2$)	$\hat{\alpha}$: No closed form $\hat{\beta} = \dfrac{\hat{\alpha}}{\frac{1}{n}\sum X_i^{-1}}$	$\begin{pmatrix} \psi'(\alpha) & -\frac{1}{\beta} \\ -\frac{1}{\beta} & \frac{\alpha}{\beta^2} \end{pmatrix}$	$\mathcal{I}(\alpha,\beta)^{-1}$
Beta $\mathrm{Beta}(a,b)$ $a > 0,\ b > 0$ *Conj. prior of Binomial.*	$f(x) = \dfrac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$ $x \in (0,1)$	$\dfrac{a}{a+b}$	$\dfrac{ab}{(a+b)^2(a+b+1)}$	$\hat{a}, \hat{b}$: No closed form	Matrix of polygammas	$\mathcal{I}(a,b)^{-1}$
Chi-Square χ^2_ν $\nu \in \mathbb{Z}^+$ *Sum of ν iid $\mathcal{N}(0,1)^2$.* $\chi^2_\nu \sim \mathrm{Gamma}(\nu/2, 1/2)$	$f(x) = \dfrac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$ $x \in (0,\infty)$	ν	2ν	$\hat{\nu}$: No closed form	$\dfrac{\psi'(\nu/2)}{4}$	$\dfrac{4}{\psi'(\nu/2)}$
Student-t t_ν $\nu > 0$	$f(x) = \dfrac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \dfrac{x^2}{\nu}\right)^{-(\nu+1)/2}$ $x \in \mathbb{R}$	0 ($\nu > 1$)	$\dfrac{\nu}{\nu-2}$ ($\nu > 2$)	$\hat{\nu}$: No closed form	Complex fn of ν	$\mathcal{I}(\nu)^{-1}$
Hypergeometric $\mathrm{HGeom}(w,b,m)$ $w, b \in \mathbb{Z}_{\geq 0}$ $m \in \{1,\ldots,w+b\}$ *Successes in m draws without replacement.*	$\mathbb{P}(X=k) = \dfrac{\binom{w}{k}\binom{b}{m-k}}{\binom{w+b}{m}}$ $k \in \{\ldots\}$	$\dfrac{mw}{w+b}$	$\dfrac{mw}{w+b}\left(1 - \dfrac{w}{w+b}\right)$ $\times \dfrac{w+b-m}{w+b-1}$	N/A	N/A	N/A
Log-Normal $\mathcal{LN}(\mu,\sigma^2)$ $\mu \in \mathbb{R},\ \sigma^2 > 0$	$f(x) = \dfrac{1}{x\sigma\sqrt{2\pi}} e^{-(\log x - \mu)^2/(2\sigma^2)}$ $x \in (0,\infty)$	$e^{\mu+\sigma^2/2}$	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$	$\hat{\mu} = \overline{\ln X_n}$ $\hat{\sigma}^2 = S^2_{\ln X_n}$	$\mathrm{diag}\left(\dfrac{1}{\sigma^2}, \dfrac{1}{2\sigma^4}\right)$	$\mathrm{diag}(\sigma^2, 2\sigma^4)$

Notes and Notation:

- $\psi'(x) = \frac{d^2}{dx^2}\ln\Gamma(x)$ is the **trigamma function**, the derivative of the digamma function.
- $X_{(1)}$ and $X_{(n)}$ denote the sample minimum and sample maximum (order statistics) of the n observations.
- $S^2_{\text{pop}} = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X_n})^2$ is the uncorrected (biased) sample variance, representing the MLE for σ^2.
- $S^2_{\ln X_n} = \frac{1}{n}\sum_{i=1}^n (\ln X_i - \overline{\ln X_n})^2$ is the uncorrected sample variance of the log-transformed data.
- $\mathcal{I}(\theta)$ is the Fisher Information evaluated for a *single* observation.
- The **Asymptotic Variance** (V) refers to the variance of the limiting distribution: $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0,V)$, where $V = \mathcal{I}(\theta)^{-1}$. i.e.: $\mathrm{Var}(\hat{\theta}_n) \approx \frac{V}{n}$.

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